

Your grade: 100%

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Next item →

1. Select if the following statements are true or false.

1 / 1 point

True	Simple linear regression involves two variables: one independent and one dependent.
False	In simple linear regression, the equation of the regression line is $y = mx + b$.
True	The slope of the regression line represents the change in the dependent variable for a one-unit change in the independent variable.
True	The intercept of the regression line represents the value of the dependent variable when the independent variable is zero.
False	The residuals in a simple linear regression model should always be normally distributed.

Response 1

- ☒ True
☐ False
☒ Nice work

Response 2

- ☐ True
☒ False
☒ Nice work
The correct form is $y = \beta_0 + \beta_1 x$

Response 3

- ☒ True
☐ False
☒ Nice work

Response 4

- ☒ True
☐ False
☒ Nice work

Response 5

- ☐ True
☒ False
☒ Nice work
Residuals should be normally distributed if the assumptions of linear regression are met, but this is not always the case.

2. The least squares method is commonly used to minimize the sum of the squared differences between the observed values and the values predicted by the regression line.

1 / 1 point

Response 1

- ☐ coefficient
☐ correlation
☐ regression line
☒ least squares
☐ residuals
☒ Nice work

3. The regression line is the line that best fits the data points in a scatter plot in simple linear regression.

1 / 1 point

Response 1

- ☐ correlation
☐ residuals
☒ regression line
☐ least squares
☐ coefficient
☒ Nice work

4. The correlation between two variables indicates how strongly they are related.

1 / 1 point

Response 1

- ☐ coefficient
☐ regression line
☒ correlation
☐ least squares
☐ residuals
☒ Nice work

5. The vertical distance between the observed data points and the values predicted by the regression line is called residuals.

1 / 1 point

Response 1

- ☐ least squares
☐ regression line
☐ correlation
☒ residuals
☐ coefficient
☒ Nice work

6. The coefficient of the regression line represents the amount by which the dependent variable changes for a one-unit change in the independent variable.

1 / 1 point

Response 1

- ☐ residuals
☐ least squares
☐ regression line
☒ coefficient
☐ correlation
☒ Nice work

7. Which of the following equations represents the general form of a simple linear regression model?

1 / 1 point

- ☐ $y = a + bx$
☐ $y = mx + c$
☐ $y = ax^2 + bx + c$
☒ $y = \beta_0 + \beta_1 x$
☒ Nice work

8. What does the coefficient of determination (R^2) indicate?

1 / 1 point

- ☐ The proportion of variance in the independent variable explained by the dependent variable.
☒ The proportion of variance in the dependent variable explained by the independent variable.
☐ The slope of the regression line.
☐ The intercept of the regression line.
☒ Nice work

9. In the context of simple linear regression, what is the purpose of the residuals?

1 / 1 point

- ☒ To quantify the discrepancy between observed and predicted values.
☐ To determine the goodness of fit of the model.
☐ To measure the slope of the regression line.
☐ To calculate the correlation coefficient.
☒ Nice work

10. Which assumption is NOT required for simple linear regression?

1 / 1 point

- ☐ Independence of residuals
☐ Homoscedasticity
☒ Normality of the independent variable
☐ Linearity
☒ Nice work

11. If the correlation coefficient between the independent and dependent variables is 0.8, what does this suggest about the relationship between the variables?

1 / 1 point

- ☐ A weak positive relationship
☐ A strong negative relationship
☐ No relationship
☒ A strong positive relationship
☒ Nice work